

Stanford University
Political Science 350D
Problem Set 3
Due April 20, 2015

1. Consider the hierarchical model where

$$\theta_i | \mu, \tau^2 \sim \text{i.i.d. Normal}(\mu, \tau^2) \quad y_{i1}, \dots, y_{in_i} | \mu, \theta, \tau^2, \sigma^2 \sim \text{i.i.d. Normal}(\theta_i, \sigma^2)$$

for $i = 1, \dots, m$, and $j = 1, \dots, n_i$.

- (a) Which do you think is bigger, $\mathbb{V}(y_{ij} | \theta_i, \sigma^2)$ or $\mathbb{V}(y_{ij} | \mu, \tau^2)$? Provide an intuitive answer without doing any calculation.

Solution: Think of i as indexing groups of observations and j as indexing units within groups. Conditioning on θ_i is equivalent to conditioning upon the group on which the observation is drawn from. On average, we expect the latter to be larger than the former because the latter only includes within group variation, while the former also includes between-group variation.

- (b) Do you think $\mathbb{C}(y_{ij}, y_{ij'} | \theta_i, \sigma^2)$ is positive, negative, or zero? Answer the same for $\mathbb{C}(y_{ij}, y_{ij'} | \mu, \tau^2)$.

Solution: Within groups, the observations are independent, so $\mathbb{C}(y_{ij}, y_{ij'} | \theta_i, \sigma^2) = 0$ if $j \neq j'$. If you don't condition on group, then observations from the same group tend to be similar (because they share the same mean θ_i) and, hence, positively correlated.

- (c) Now compute each of the following quantities and compare to your answers in parts (a) and (b).

i. $\mathbb{V}(y_{ij} | \theta_i, \sigma^2)$

Solution: $\mathbb{V}(y_{ij} | \theta_i, \sigma^2) = \sigma^2$.

ii. $\mathbb{V}(\bar{y}_{ij} | \theta_i, \sigma^2)$

Solution: $\mathbb{V}(\bar{y}_{ij} | \theta_i, \sigma^2) = \sigma^2 / n_i$.

iii. $\mathbb{C}(y_{ij}, y_{ij'} | \theta_i, \sigma^2) \ (j \neq j')$

Solution: $\mathbb{C}(y_{ij}, y_{ij'} | \theta_i, \sigma^2) = 0$ if $j \neq j'$.

iv. $\mathbb{V}(y_{ij} | \mu, \tau^2)$

Solution:

$$\begin{aligned} \mathbb{V}(y_{ij} | \mu, \tau^2) &= \mathbb{E}[\mathbb{V}(y_{ij} | \theta_i, \sigma^2, \mu, \tau^2) | \mu, \tau^2] + \mathbb{V}[\mathbb{E}(y_{ij} | \theta_i, \sigma^2, \mu, \tau^2) | \mu, \tau^2] \\ &= \mathbb{E}(\sigma^2 | \mu, \tau^2) + \mathbb{V}(\theta_i | \mu, \tau^2) = \mathbb{E}(\sigma^2 | \mu, \tau^2) + \tau^2. \end{aligned}$$

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v. $\mathbb{V}(\bar{y}_i | \mu, \tau^2)$

Solution:

$$\begin{aligned}\mathbb{V}(\bar{y}_i | \mu, \tau^2) &= \frac{\sum_{j=1}^{n_i} \mathbb{V}(y_{ij} | \mu, \tau^2) + \sum_{j \neq j'} \mathbb{C}(y_{ij}, y_{ij'} | \mu, \tau^2)}{n_i^2} \\ &= \frac{n_i [\mathbb{E}(\sigma^2 | \mu, \sigma^2) + \tau^2] + n_i(n_i - 1)\tau^2}{n_i^2} = \frac{\mathbb{E}(\sigma^2 | \mu, \tau^2) + n_i \tau^2}{n_i}\end{aligned}$$

vi. $\mathbb{C}(y_{ij}, y_{ij'} | \mu, \tau^2)$ ($j \neq j'$)

Solution:

$$\begin{aligned}\mathbb{C}(y_{ij}, y_{ij'} | \mu, \tau^2) &= \mathbb{E}[\mathbb{C}(y_{ij}, y_{ij'} | \theta^2, \sigma^2) | \mu, \sigma^2] + \mathbb{C}[\mathbb{E}(y_{ij} | \theta_i, \sigma^2), \mathbb{E}(y_{ij'} | \theta_i, \sigma^2) | \mu, \tau^2] \\ &= 0 + \mathbb{C}(\theta_i, \theta_i | \mu, \tau^2) = \tau^2.\end{aligned}$$

(d) Show that if we have a prior $p(\mu)$ for μ , then

$$p(\mu | \theta_1, \dots, \theta_m, \sigma^2, \tau^2, y) = p(\mu | \theta_1, \dots, \theta_m, \tau^2).$$

Provide a verbal interpretation of this result.

Solution: The prior is $p(\theta, \sigma^2, \mu, \tau^2)$, while the likelihood is

$$p(y | \theta, \sigma^2, \mu, \tau^2) = \prod_{i=1}^m \prod_{j=1}^{n_i} p(y_{ij} | \theta_i, \sigma^2)$$

so

$$p(\mu, \theta, \sigma^2, \tau^2, y) = p(\mu, \theta, \sigma^2, \tau^2) \prod_{i=1}^m \prod_{j=1}^{n_i} p(y_{ij} | \theta_i, \sigma^2).$$

It follows that

$$\begin{aligned}p(\mu | \theta, \sigma^2, \tau^2, y) &= \frac{p(\mu, \theta, \sigma^2, \tau^2, y)}{\int p(\mu, \theta, \sigma^2, \tau^2, y) d\mu} \\ &= \frac{p(\mu, \theta, \sigma^2, \tau^2) \prod_{i=1}^m \prod_{j=1}^{n_i} p(y_{ij} | \theta_i, \sigma^2)}{\prod_{i=1}^m \prod_{j=1}^{n_i} p(y_{ij} | \theta_i, \sigma^2) \int p(\mu, \theta, \sigma^2, \tau^2) d\mu} \\ &= \frac{p(\mu, \theta, \sigma^2, \tau^2)}{\int p(\mu, \theta, \sigma^2, \tau^2) d\mu} = p(\mu | \theta, \sigma^2, \tau^2).\end{aligned}$$

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In words, there is no information in the observations y about the hyperparameter μ given the θ and the other hyperparameters. The observations y are informative about θ , but, given θ , are conditionally independent of μ .

2. Use the model in question 1 with the following data where $m = 5$:

i	n_i	y_{i1}	y_{i2}	y_{i3}	y_{i4}	y_{i5}	y_{i6}
1	2	1.75	-0.52				
2	4	-1.77	-1.46	-3.21	0.51		
3	4	1.97	-2.35	-0.85	0.06		
4	4	1.11	-0.80	1.34	0.64		
5	6	1.87	1.44	1.38	2.45	0.73	1.44

- (a) Estimate θ_i using the sample mean $\bar{y}_i = n_i^{-1} \sum_{j=1}^{n_i} y_{ij}$.

Solution:

```
> data
$y
 [1]  1.75 -0.52 -1.77 -1.46 -3.21  0.51  1.97 -2.35 -0.85
[10]  0.06  1.11 -0.80  1.34  0.64  1.87  1.44  1.38  2.45
[19]  0.73  1.44

$group
 [1] 1 1 2 2 2 2 3 3 3 3 4 4 4 4 5 5 5 5 5 5

$J
 [1] 5

$N
 [1] 20

> means <- with(data, by(y, group, mean))
> as.numeric(means)
 [1]  0.615000 -1.482500 -0.292500  0.572500  1.551667
```

- (b) Using a Uniform($-6, 6$) prior for μ and independent Scaled Inverse Chi-squared($1, 1$) priors for σ^2 and τ^2 , estimate the posterior means for the parameters using JAGS. Call the posterior means $\hat{\theta}_i$ for $i = 1, \dots, 5$.

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Solution: A Scaled Inverse Chi-squared(1, 1) distribution is equivalent to an Inverse Gamma(1/2, 1/2) distribution. Here are the posterior means computed using JAGS:

```
> library(rjags)
> model <- "model {
+   for (i in 1:N) {
+     y[i] ~ dnorm(theta[ group[i] ], sigma2.inv)
+   }
+   for (j in 1:J) {
+     theta[j] ~ dnorm(mu, tau2.inv)
+   }
+   mu ~ dunif(-6, 6)
+   sigma2.inv ~ dgamma(0.5, 0.5)
+   tau2.inv ~ dgamma(0.5, 0.5)
+   sigma <- 1 / sqrt(sigma2.inv)
+   tau <- 1 / sqrt(tau2.inv)
+ }"
>
> sampler <- jags.model(textConnection(model), data=data)
|+++++| 100%
> update(sampler, 1000)
|*****| 100%
> samples <- jags.samples(sampler, n.iter=10000,
+   variable.names=c("mu", "sigma", "tau", "theta"))
|*****| 100%
> samples
$mu
marray:
[1] 0.2036414

Marginalizing over: iteration(10000),chain(1)

$sigma
marray:
[1] 1.301172

Marginalizing over: iteration(10000),chain(1)

$tau
marray:
[1] 1.163716
```

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```
Marginalizing over: iteration(10000),chain(1)

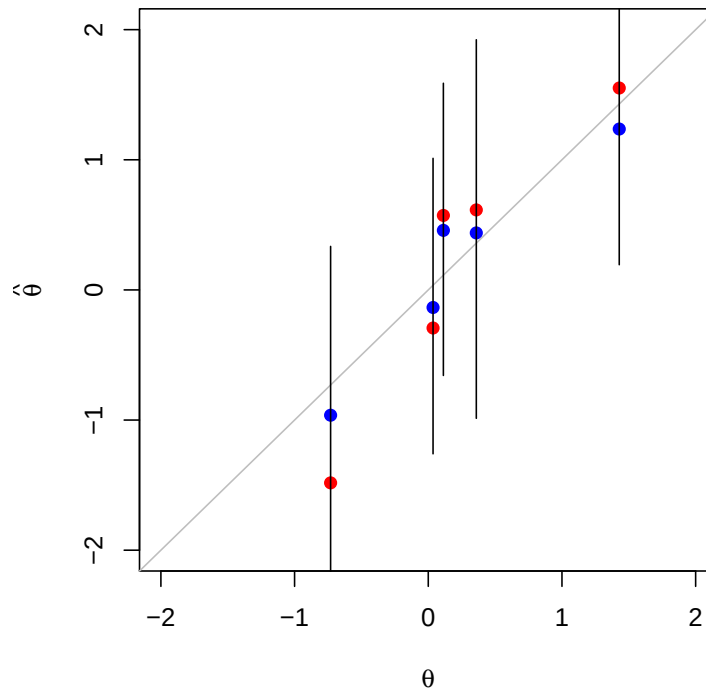
$theta
marray:
[1] 0.4520188 -0.9706348 -0.1419024 0.4604863 1.2501051

Marginalizing over: iteration(10000),chain(1)
```

- (c) These data were simulated using $\theta = (0.36, -0.73, 0.04, 0.11, 1.43)$. Compare the sample means $\bar{y}_i$ and the posterior means $\hat{\theta}_i$ to θ_i . (Use a scatterplot.)
- (d) Draw 95% posterior credible intervals onto the scatterplot in part (c).

Solution:

Comparison of Hierarchical Bayes with Least Squares



Here is the R code to produce the plot:

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```
theta.hat <- summary(samples$theta, mean)$stat
lo <- summary(samples$theta, quantile, prob=.025)$stat
hi <- summary(samples$theta, quantile, prob=.975)$stat
plot(theta, theta.hat, pch=19, col="blue", xlim=c(-2,2),
      xlab=expression(theta), ylim=c(-2,2), ylab=expression(hat(theta)),
      main="Comparison of Hierarchical Bayes with Least Squares")
points(theta, means, pch=19, col="red")
abline(0, 1, col="grey")
for (i in 1:length(theta)) lines(x=rep(theta[i], 2), y=c(lo[i], hi[i]))
```

Note: JAGS (unlike Stan) does not accept ragged arrays. I would suggest that you make $y = c(1.75, -0.52, -1.77, \dots)$ into a vector of 20 elements. You can then create a variable `group <- rep(1:5, c(2,4,4,4,6))` so that $y[i] \sim \text{dnorm}(\theta_{\text{group}[i]}, \text{sigma2inv})$ for $i = 1, \dots, 20$.